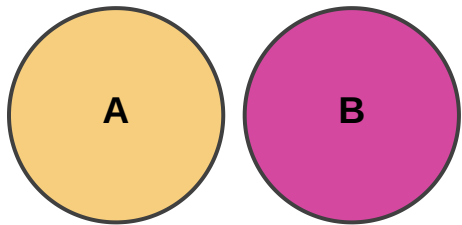


A and B disjoint

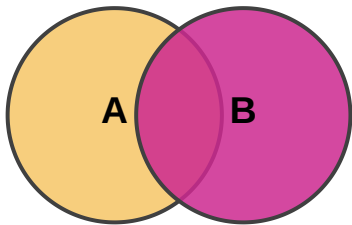
$$P(A \cup B) = P(A) + P(B)$$



$$A \cap B = \emptyset \text{ (disjoint)}$$

A and B not disjoint

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$$A \cap B \neq \emptyset \text{ (not disjoint)}$$